

Grade Eight Science - Body Systems Unit 1

MYP Criterion A Practice Questions

- This document contains twelve practice questions for the upcoming Body Systems assessment
- All questions are based on MYP Criterion A (See below)
- There are three practice questions from each level; one for each strand
- The MYP Command Terms used in the questions are provided below

MYP Science Assessment Criterion A: Knowing and Understanding

Achievement Level	Level Descriptor
0	The student does not reach a standard indicated by any of the descriptors below
1-2	The student is able to: • recall scientific knowledge • apply scientific knowledge and understanding to suggest solutions to problems set in familiar situations • apply information to make judgments
3-4	The student is able to: • state scientific knowledge • apply scientific knowledge and understanding to solve problems set in familiar situations • apply information to make scientifically supported judgments.
5-6	The student is able to: • outline scientific knowledge • apply scientific knowledge and understanding to solve problems set in familiar situations and suggest solutions to problems set in unfamiliar situations • interpret information to make scientifically supported judgments
7-8	The student is able to: • describe scientific knowledge • apply scientific knowledge and understanding to solve problems set in familiar and unfamiliar situations • analyze information to make scientifically supported judgments

MYP Command Terms Used in the Questions

State: Give a specific name, value, or other brief answer without explanation of calculator	Outline: Give a brief account or summary
Apply: Use knowledge and understanding in response to a given situation	Describe: Give a detailed account or picture of a situation, event, pattern, or process
Recall: Remember or recognize from prior learning experience	Suggest: Propose a solution, hypothesis, or other possible answer
Analyze: To identify parts and relationships, and to interpret information to reach conclusions	Interpret: Use knowledge and understanding to recognize trends and draw conclusions from given information

LEVELED BODY SYSTEMS PRACTICE QUESTIONS - CRITERION A

Level 1-2 Practice Questions

1. **Recall** that the nervous system is made up of the:
- A. Heart, brain, and spinal cord
 - B. Nerves, brain, and spinal cord
 - C. Nerves, brain, spinal cord, and heart
 - D. Nerves and Brain
2. A person went to the hospital with trouble breathing. **Apply** scientific knowledge and understanding to **suggest** which part of the human anatomy, from the list below, is likely damaged.
- A. Trachea
 - B. Esophagus
 - C. Stomach
 - D. Pancreas
 - E. None of the above
 - F. All of the above
3. A student left school with stomach pain. When examined, the doctor found an ulcer. This means that acid burned through the stomach lining. **Apply** information to choose which item, from the list below, likely caused the ulcer. Was it caused by:
- A. Folds in the stomach
 - B. Peristalsis
 - C. Valves not working properly
 - D. Not enough mucus

Level 3-4 Practice Questions

1. **State** the difference between an artery and a vein

2. During a lesson on the nervous system, a student hits their knee with a reflex hammer. They were surprised when nothing happened, as they expected their leg to lift upward. **Apply** your scientific knowledge and understanding to help the student understand why they did not get the result they expected.

Level 5-6 Practice Questions:

1. **Apply** information learned in class to make a scientifically supported judgement about why the intestines are the longest part of the human digestive system

2. At high altitudes the air is said to be "thin," meaning that it has less oxygen. **Apply** scientific knowledge and understanding to **suggest one way** the human body could be adapted to make living at high altitudes easier

3. A coach tells a team of Sprinters that, because they want to leave the start as fast as possible, they should rely on visual cues, rather than sound cues at the start of a race.

Table 1. Reaction Times of Different Stimulus

Type of Stimulus	Reaction Time (milliseconds)
Visual (Seeing a light)	250
Auditory (Hearing a beep)	270
Tactile (feeling a tap)	200

Interpret data from Table 1 above to make a scientifically supported judgement about:

- A. **Describe** the body system is being utilized in the coaches discussion
- B. If the coach is giving information that makes sense to the Sprinters

Answer:

Level 7-8 Practice Questions:

1. **Describe** scientific knowledge by giving two similarities and two differences between the Digestive and Circulatory systems

Two Similarities

Similarity Number One	Similarity Number Two

Two Differences

Difference Number One	Difference Number Two

2. Smoking is a global problem that impacts different body systems. Smoking is a process where chemicals are inhaled into the lungs. Smoking can cause lung and pharynx cancer and it makes veins and arteries become more narrow. It can also cause breathing problems.

After reading the paragraph above, **apply** scientific knowledge and understanding to decide which three body systems studied in class are impacted by smoking. Use specific pieces of information from the paragraph to support your answer.

Body System	Specific Information to Support

3. **Analyze** information from the paragraph in question 2 above to decide
- A. What factor might be causing veins and arteries to become more narrow in smokers
 - B. Using scientific knowledge learned in class **describe** one problem that you think could be caused by narrow veins and arteries